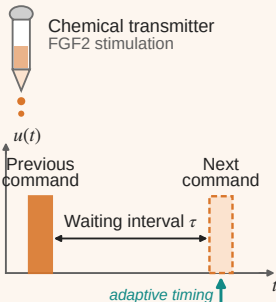
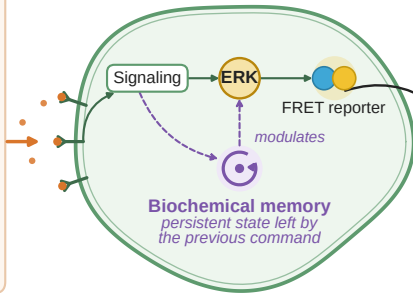


Repeated chemical commands



Living cellular receiver (PC-12 cell)



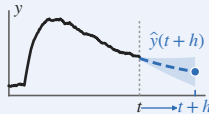
Observed response $y(t)$



1 Future-response prediction

Exp. data + mech. model

Can we predict the future response?



Which information helps?

Current state

Temporal history

Mechanistic information

2 Adaptive receiver reuse

Mech. model (simulated)

When can the receiver be reused?



$\hat{R}_t \geq c$
 $\Rightarrow$ issue next command

Adaptive reuse: issue the next command once $\hat{R}_t \geq c$